Climate Analysis Report of Major Indian Cities (2024–2025)

1. Introduction

This report presents a comprehensive climate analysis of major Indian cities during the period **2024–2025**, based on a dataset containing **7,310 records** across **13 meteorological variables**.

The objective of this study is to extract meaningful insights from climate data, focusing on:

- Temperature patterns
- Rainfall and humidity relationships
- Wind speed distribution
- Air Quality Index (AQI) trends

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2. Dataset Overview

The dataset includes the following key parameters:

- Temperature (Max & Average)
- Humidity (%)
- Rainfall (mm)
- Wind Speed (km/h)
- Cloud Cover (%)
- Air Quality Index (AQI)
- AQI Category
- Date (time-series analysis)

Key Observations

- Total records: **7,310**
- Total features: **13**
- Missing values: **None**

This makes the dataset **highly reliable and suitable for analysis**.

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3. Key City-Level Insights

3.1 Temperature Analysis

- **Kolkata** recorded the **highest average maximum temperature**
- Indicates strong heat concentration in eastern regions

3.2 Air Quality Analysis

- **Mumbai** had the **lowest average AQI**
- Implies comparatively **cleaner air quality**

3.3 Cloud Cover

- **Lucknow** showed the **highest average cloud cover**
- Suggests more overcast conditions compared to other cities

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4. Wind Speed Analysis

- **Mumbai** experienced the **highest average wind speeds**
- Inland cities like **Bhopal and Lucknow** also showed strong wind patterns
- **Delhi and Bengaluru** recorded **lower wind speeds (~13.2 km/h)**

Insight

Lower wind speeds → poor pollutant dispersion
This helps explain **Delhi's air quality issues**

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5. Air Quality Distribution (AQI Categories)

The AQI was categorized into:

- Good
- Satisfactory
- Moderate
- Poor
- Very Poor
- Severe

Insight

- Majority of data falls into **Moderate to Poor categories**
- Indicates **frequent exposure to unhealthy air conditions**

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6. Rainfall vs Humidity Analysis

Key Patterns

- **Ahmedabad & Kolkata**:

    - High rainfall + high humidity
    - Located in **upper-right cluster**
- **Bengaluru**:
    - Lowest rainfall
    - Moderate humidity (~62%)
- **Chennai & Lucknow**:
    - Lower humidity levels compared to others

Conclusion

- Positive correlation exists between **rainfall and humidity**
- Coastal and eastern cities show **higher moisture retention**

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7. Time-Series Temperature Trends

Using monthly aggregation:

- Clear **seasonal variation** observed
- Temperature peaks during **summer months**
- Drops during **monsoon and winter**

Insight

- Confirms **predictable seasonal climate cycles**
- Useful for:
    - Agriculture planning
    - Energy demand forecasting

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8. Overall Findings

Climate Patterns

- Eastern cities → hotter & more humid
- Coastal cities → better air quality + higher winds
- Northern cities → lower wind speed + worse AQI

Environmental Insights

- Wind speed plays a **critical role in pollution dispersion**
- Urban air quality is strongly influenced by:
    - Weather conditions
    - Geographic location

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9. Conclusion

This study successfully transforms raw climate data into actionable insights:

- Identifies **city-wise climate behavior**
- Highlights **air quality concerns**
- Establishes relationships between:
    - Rainfall & humidity
    - Wind speed & pollution